

Discovering Molecular Substructures from Mass Spectra with Neural Topic Models

Abstract

Substructure discovery from mass spectra seeks recurring fragmentation patterns and an interpretable account of how they combine within individual spectra. We present Contextual Sparse ETM, an adaptation of the published Embedded Topic Model that retains its embedding-based decoder and Gaussian inference network. Three enhancements balance fragment and neutral-loss probabilities, use whole-spectrum evidence to guide inference, and allow exactly zero topic weights through 1.5-entmax. Across three training runs evaluated on 3,889 MSⁿLib validation spectra, the selected model produces 390.7 ± 2.1 motifs meeting the specified chemical screening criterion, compared with 351.7 ± 7.0 for Tomotopy LDA. Completion negative log-likelihood is lower, but mean chemical overlap among evaluable motifs is also lower. The median entropy-effective number of topics per spectrum is 3.72 ± 0.03 . Matched-background analysis finds similar small excess chemical overlap in both models, and much of the additional evaluable neural inventory has single-compound support. Cross-model matching identifies shared spectral patterns, with stronger agreement in fragments than losses and limited agreement in supporting compounds. These findings support a broader candidate inventory with compact per-spectrum explanations, rather than more confirmed substructures. Because the validation data also informed model selection, independent evaluation and structural follow-up remain necessary.

1 Introduction

Untargeted metabolomics uses mass spectrometry to survey small molecules without restricting an experiment to a predefined list of compounds. Tandem mass spectrometry (MS/MS) adds structural information by fragmenting an isolated precursor ion and recording the mass-to-charge ratios (m/z) and intensities of the resulting ions. Relating those patterns to molecular structures remains difficult: different molecules can share fragments, and the spectrum of one molecule can change with acquisition conditions. Reference-library matching is valuable when suitable standards are available, but incomplete library coverage motivates complementary ways to interpret unidentified spectra [7, 1].

Substructure discovery asks a different question from complete molecular identification. Instead of selecting one molecule for each spectrum, it seeks recurring evidence for chemical building blocks across many spectra. MS2LDA casts this as topic modeling: spectra are documents, binned fragments and neutral losses are words, and recurring word distributions are *Mass2Motifs* [1]. A spectrum can contain several motifs, and a motif can occur in chemically different molecules. This is useful for relating compounds whose complete spectra need not be close matches. Each pattern suggests a possible chemical interpretation, but does not by itself identify a unique substructure or establish a fragmentation pathway.

MS2LDA 2.0 makes this workflow substantially more practical through the Tomotopy topic-modeling backend, improved software interfaces and automated Mass2Motif Annotation Guidance (MAG) [2]. These advances address computational and annotation bottlenecks; they do not make every learned topic chemically informative. Its library benchmark includes motifs that cannot be retained after annotation refinement, lack associated molecules, or have weak structural agreement. We therefore focus on improving the learned motifs and their assignments to spectra, while retaining these software and annotation advances.

The existing LDA model learns a topic-word probability table and already uses both fragment/loss co-occurrence and mixtures of topics within a spectrum. It does not explicitly share statistical strength through continuous spectral-word embeddings or learn a reusable neural inference function. LDA can nevertheless represent both concentrated mixtures and contextual dependence. We investigate whether an embedding-based neural alternative can increase the number of motifs meeting our chemical screening criterion while maintaining compact per-spectrum mixtures. We do not attempt automatic selection of the number of topics, removal of reference-library bias, or tokenization-free spectral modeling.

We build on the published Embedded Topic Model (ETM) [4]. ETM represents each topic by a vector in the same space as the spectral-word embeddings. The topic-word inner products define a distribution over words; a multilayer perceptron (MLP) infers how topics combine within each spectrum. We adapt this model in three ways:

1. *Balanced fragment and loss emissions.* Give the two spectral channels equal probability mass within each topic, so they do not compete for the topic’s total mass.

2. *Whole-spectrum contextual evidence.* Match a word together with its surrounding spectrum to the topic embeddings, and use the pooled evidence to correct the encoder’s Gaussian mean.
3. *Sparse topic mixtures.* Replace softmax with 1.5-entmax, allowing weakly supported topics to receive exactly zero weight.

Figure 2 gives a worked illustration of each change. The model retains ETM’s shared topics, Gaussian latent representation and variational training framework.

We evaluate the model in two settings: controlled synthetic spectra, where the planted motifs are known, and real reference spectra, where Tomotopy LDA is the primary comparator. Plain ETM provides a secondary baseline for assessing the combined enhancements. Our contribution is a spectral adaptation of an established neural topic model, evaluated for both motif recovery and practical substructure discovery.

2 Related work

Spectral topics and reusable motif annotations. LDA represents documents as mixtures of latent topics, with a probability distribution over words for each topic [3]. The original MS2LDA transfers that construction to fragments and neutral losses, enabling unsupervised discovery of patterns shared across spectra [1]. Subsequent work introduced MotifDB to store, share and reuse annotated Mass2Motifs, linking discovery with community annotation [14]. MS2LDA 2.0 extends the practical workflow with scalable fitting and MAG [2]. We retain its motif interpretation and annotation framework so that the main comparison concerns the topic model, not replacement of the entire discovery pipeline.

Learning relationships between spectral features. Skip-gram with negative sampling learns word coordinates from co-occurrence [6]. Spec2Vec applies this idea to mass spectra: fragments and losses receive embeddings that support spectrum-level similarity [7]. This motivates using learned spectral relationships rather than treating every vocabulary entry as unrelated, although an embedding distance is not itself proof of a shared substructure. Our ETM uses word embeddings trained only on its training spectra. The pretrained Spec2Vec model used by MAG is a separate annotation asset, not the ETM representation. More recently, DreaMS learns representations directly from large collections of spectra through self-supervised training [16]. Such models could provide future spectrum encoders; they are not used in these experiments.

Neural probabilistic topic models. Autoencoding variational inference replaces repeated document-specific optimization with a learned recognition network. The AVITM framework also introduced ProdLDA, which changes the observation combination from LDA’s mixture to a product of experts [5]. ETM instead uses embedding inner products to parameterize topic–word distributions while retaining an additive topic mixture [4]. This makes ETM a particularly transparent base for Mass2Motifs: each topic still has a normalized distribution over identifiable fragment/loss words. We therefore retain ETM’s Gaussian latent variables, two-hidden-layer MLP and variational training objective. The official Pyro tutorial illustrates a ProdLDA implementation [17]; our experiments use neither that implementation nor its product-of-experts decoder.

Contextual inference and sparse probability maps. Topic-aware neural inference predates this work. For example, TAN-NTM uses topic attention over contextual sequence representations before variational inference [15]. Our permutation-invariant inference path combines each word embedding with the mean embedding of the whole spectrum, compares the resulting vector with the decoder’s topic vectors, and adds the pooled evidence to the Gaussian posterior mean. There is no recurrent sequence encoder or second topic dictionary. The entmax family provides published probability mappings that can produce exact zeros [8]. We use its 1.5 form to modify ETM’s topic mixtures. Both attention and sparse probability maps are established methods; here we combine them within ETM to model spectral motifs.

3 Materials and methods

We first describe the spectral input and base ETM, then explain the three enhancements and how they fit together. The main text retains the model’s probabilistic structure and worked examples. Full notation, mathematical details and reproducible protocols are in the accompanying [Supplementary Information](#).

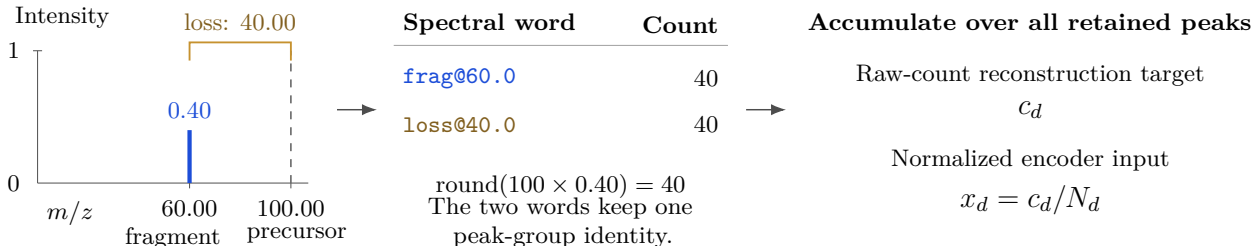
3.1 Spectra, spectral words and evaluation data

We use the positive-ion MSⁿLib benchmark distributed with MS2LDA 2.0 [9, 10]. Filtering leaves 38,888 of 41,568 input spectra. Each retained peak contributes a fragment word and, where applicable, a precursor-minus-fragment neutral-loss word. Masses are rounded to two decimal places; a peak with relative intensity I

contributes $\text{round}(100I)$ copies of each word. These are intensity-derived pseudo-counts, rather than independent ion counts. Figure 1 gives a numerical example. The model reconstructs the count vector c_d for spectrum d , while the encoder receives its normalized version $x_d = c_d / \sum_w c_{dw}$.

Molecules are grouped by connectivity and scaffold before splitting, giving 27,222 training, 3,889 validation and 7,777 reserved test spectra, with no compound or split-group overlap. The vocabulary contains 21,233 words occurring in at least three training spectra. We learn 48-dimensional word embeddings from training spectra using skip-gram with negative sampling [6] and hold them fixed during topic-model fitting. Neither validation spectra nor chemical labels train the vocabulary, embeddings or model weights. The reported results use development-validation data that also informed model selection; the reserved test set is not used for the comparisons reported here. It was examined during earlier development and is therefore not an untouched confirmation set. Filtering, tokenization and splitting details are in Supplementary Section S1.

(a) One retained peak becomes two spectral words



(b) Structure groups stay within one partition

38,888 filtered spectra; no shared compound or split group between partitions

Training: 27,222	Validation: 3,889	Reserved test: 7,777
Fit vocabulary, word vectors and model. $V = 21,233$; $E = 48$	Compare fitted models; results reported here.	Excluded from these reported comparisons.

Validation uses the vocabulary and word vectors fitted on training spectra.

Figure 1: Spectral representation and evaluation boundaries. (a) An illustrative peak with normalized intensity 0.40 contributes paired fragment and neutral-loss pseudo-counts when both words occur in the training vocabulary. Accumulated raw counts train reconstruction; normalized counts enter inference. The dashed precursor marker indicates mass, not intensity. (b) Structure-grouped partitions have different roles; validation does not fit the vocabulary or word embeddings.

Controlled synthetic data provide a separate recovery experiment. Each realization contains 18 planted motifs, one to three active motifs per spectrum, and 800 training and 160 validation spectra. Shared anchors, background peaks and intensity variation make recovery nontrivial. We fit 36 or 128 topics and assess recovery against the planted words and mixtures. The generator and evaluation targets are described in Supplementary Section S1; they test statistical recovery, not a mechanism of real molecular fragmentation.

3.2 Base model: the published ETM

ETM explains each spectrum as a mixture of shared topics. A topic is a probability distribution over spectral words; its weight describes how strongly it contributes to a spectrum. We use the published variant with pre-fitted word embeddings [4]. Let ρ_w be the fixed vector for word w , α_k the learned vector for topic k , V the vocabulary size, and K the number of topics. The topic’s word probabilities are

$$\beta_{kw}^{\text{ETM}} = \frac{\exp(\alpha_k^\top \rho_w)}{\sum_{v=1}^V \exp(\alpha_k^\top \rho_v)}. \quad (1)$$

The inner product measures topic–word compatibility. Words with similar embeddings interact similarly with a topic vector, sharing information across the vocabulary.

The generative model draws Gaussian topic scores z_d , converts them to mixture weights θ_d , and draws words from the weighted topic distributions:

$$z_d \sim \mathcal{N}(0, I_K), \quad \theta_d = \text{softmax}(z_d), \quad p(w \mid z_d) = \sum_{k=1}^K \theta_{dk} \beta_{kw}^{\text{ETM}}. \quad (2)$$

For example, weights $(0.7, 0.3)$ and a fragment with probabilities 0.4 and 0.1 under the two topics give predicted probability $0.7(0.4) + 0.3(0.1) = 0.31$. The enhanced model retains this additive interpretation.

For an observed spectrum, the topic scores are unknown. A two-hidden-layer MLP predicts a Gaussian mean $\mu_\phi(x_d)$ and log variance $\ell_\phi(x_d)$:

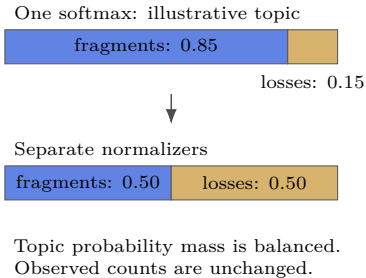
$$q_\phi(z_d | x_d) = \mathcal{N}(\mu_\phi(x_d), \text{diag}(\exp \ell_\phi(x_d))). \quad (3)$$

The same encoder estimates the mixture for every spectrum, including a new one. Training jointly fits topic vectors and encoder parameters, rewarding reconstruction of observed counts while regularizing the inferred Gaussian toward the standard-normal prior. This is ETM’s variational training framework. Supplementary Section S2 gives the full objective, sampling calculation and notation.

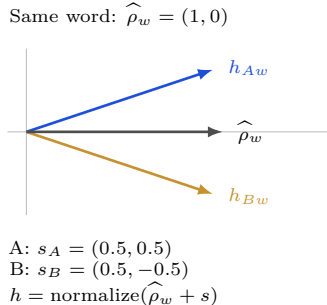
3.3 Enhancement 1: balance fragment and loss probabilities

The first change is to the decoder. Fragments and losses give complementary views of a peak, but base ETM puts all words into one softmax. A topic could therefore assign 85% of its probability to fragments and only 15% to losses. We normalize the words separately within each channel and give each channel 50% of every topic’s probability (Figure 2a). The model still learns which words matter within each channel.

(a) Channel probability mass



(b) Whole-spectrum context



(c) Sparse topic proportions

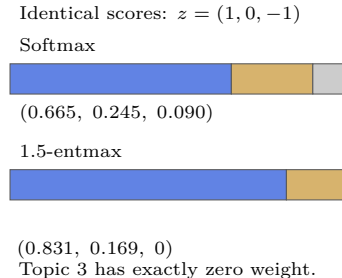


Figure 2: Illustrations of the three enhancements, not measured model outcomes. In (a), bar length is total probability in each spectral channel. In (b), a shared word vector is tilted by two different spectrum means, with context strength one. The resulting unit directions are approximately $h_{Aw} = (0.949, 0.316)$ and $h_{Bw} = (0.949, -0.316)$; the axes are embedding coordinates. In (c), colored segments are topic 1, topic 2 and topic 3, in that order; each complete bar has unit mass.

The resulting β remains a topic–word probability table and introduces no additional learned parameter. Equal channel mass is a modeling assumption; it does not force equal observed counts or equal gradient contributions. The complete emission equation and its likelihood interpretation are in Supplementary Section S2.2.

3.4 Enhancement 2: use the surrounding spectrum to guide inference

The second change is to the Gaussian mean inferred by the encoder. A fragment can support different topics depending on which fragments and losses accompany it. Although the base MLP already sees the full spectrum, we add an explicit route that uses the decoder’s embedding geometry to guide inference.

This route has four steps. We compute the intensity-weighted mean of the spectrum’s unit word vectors; add a learned multiple of that mean to each word vector and normalize its direction; match that direction to its two closest topic directions; then pool these matches with the observed word weights. A centered log transformation converts the pooled evidence into a mean correction o_d :

$$m_d = \mu_\phi(x_d) + o_d. \quad (4)$$

The posterior remains Gaussian, with corrected mean m_d and the MLP’s predicted variance. Uniform evidence gives zero correction. Top-2 matching is a restriction per word, not a limit of two topics per spectrum. The context pathway adds one learned strength scalar; it shares the decoder’s topic vectors.

Figure 2b illustrates the idea. The same fragment vector $(1, 0)$, combined with spectrum means $(0.5, 0.5)$ and $(0.5, -0.5)$ at context strength one, points in different directions. Its stored vector is unchanged, but its match to topics can differ. These illustrative coordinates explain the calculation; they do not assign a chemical structure. The exact matching, pooling and mean-correction formulas are in Supplementary Section S2.3.

3.5 Enhancement 3: let weak topic weights become zero

The third change maps Gaussian scores to sparse topic weights using $\theta_d = \text{entmax}_{1.5}(z_d)$ [8]. Softmax assigns every topic a positive weight. For scores $(1, 0, -1)$, it gives approximately $(0.665, 0.245, 0.090)$; 1.5-entmax instead gives $(0.831, 0.169, 0)$ (Figure 2c). The weakest topic has exactly zero weight, without a post-hoc cutoff or a fixed number of active topics.

The intended benefit is a compact explanation of each spectrum while allowing different spectra to use different topics. This changes the generative model: the latent scores retain their Gaussian prior, but their mixture weights can now include exact zeros while remaining nonnegative and summing to one. The mapping’s threshold formula and mathematical properties are in Supplementary Section S2.4.

3.6 Putting the three enhancements together

The complete model uses balanced topic distributions β , the corrected posterior mean m_d , and entmax topic weights. Enhancements 1 and 3 change the generator; enhancement 2 changes inference. Fixed word embeddings, learned topic vectors, the two-layer MLP and the Gaussian prior remain in place. Figure 3 shows how a few topics combine to explain words.

The training objective retains ETM’s reconstruction-minus-regularization form:

$$\mathcal{L}_d = \underbrace{\mathbb{E}_q[\log p(c_d | z_d)]}_{\text{reconstruct observed counts}} - \underbrace{\text{KL}(q(z_d | x_d) \parallel \mathcal{N}(0, I_K))}_{\text{regularize the inferred Gaussian}}. \quad (5)$$

Here q uses the corrected mean m_d . The Kullback–Leibler (KL) divergence measures departure from the prior. Training samples Gaussian scores; deterministic inference uses $\hat{\theta}_d = \text{entmax}_{1.5}(m_d)$, an estimate at the posterior mean rather than an average over posterior samples. The expanded likelihood, analytic KL, numerical safeguards and optimization details are in Supplementary Section S2.5.

(a) From normalized counts to latent topic scores

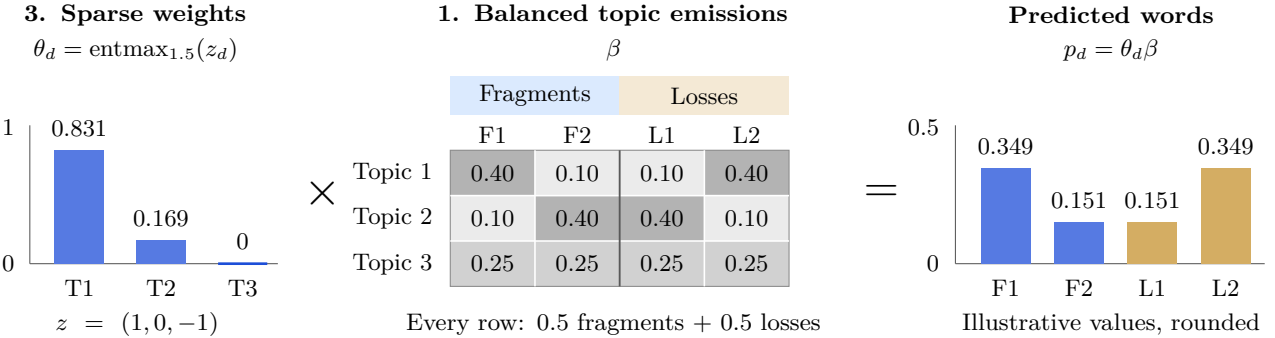
$$m_d = \underbrace{\mu_\phi(x_d)}_{\text{ETM MLP}} + \underbrace{o_d}_{\text{2. Whole-spectrum evidence}} \quad \ell_d = \ell_\phi(x_d)$$

$$q(z_d | x_d) = \mathcal{N}(m_d, \text{diag}(\exp \ell_d))$$

Training: draw scores z_d from this Gaussian

Inference: $z_d = m_d$

(b) Combine a few topics to predict spectral words



Shared embeddings: fixed word vectors ρ and learned topic vectors α .
Context matches their unit directions; the decoder uses their inner products.

Training objective: raw-count reconstruction minus Gaussian KL (Equation 5).

Figure 3: How the selected model infers and explains a spectrum. (a) The ETM encoder supplies Gaussian parameters; whole-spectrum evidence corrects the mean. Training samples latent scores; deterministic inference uses their mean. (b) Sparse topic weights combine balanced topic–word distributions into word probabilities. The example illustrates this computation, not experimental performance. T1–T3 identify three topics; F1/F2 and L1/L2 denote illustrative fragment and loss words. Numbered labels identify the three enhancements.

3.7 Comparators and experimental design

Tomotopy LDA, the production MS2LDA backend, is the primary comparator. It already models fragment/loss co-occurrence and concentrated mixtures. The neural model differs in its embedding-constrained topic probabilities and reusable inference network. A component-by-component comparison is provided in Supplementary Table S2.

All real-data models use the same training spectra, vocabulary, 1,000 topics, validation split and chemical scoring procedure. Each is summarized across three fits. The neural models share fixed embeddings and an MLP with two 800-unit hidden layers, trained for 120 epochs. Plain ETM is the secondary comparator for the enhancement package; its batch size is 256 rather than the selected model’s 200. Computational budgets are not matched to Tomotopy, and this comparison does not isolate the effect of each enhancement. Settings and inference protocols are in Supplementary Section S3.

Real-data means and sample SDs describe variation between fitted runs on one fixed partition; synthetic SDs also include changes in generated data and embeddings across three realizations. The original plain-ETM fit used WSL2 and its repetitions native Linux, with matching scientific package versions. The SDs are not confidence intervals over new chemical populations or pure seed-only effects. Exact random states and environment records are in Supplementary Section S8.2.

3.8 What the evaluation measures

The metrics answer different questions and are not combined into one score. Their full definitions and numerical conventions are in Supplementary Section S4.

Recovery, prediction and mixture compactness. Synthetic recovery compares fitted and planted word distributions after one-to-one topic matching. We report word and mixture cosine similarities (higher is better) and how many of the 18 motifs attain word cosine at least 0.50. For nonnegative word or mixture vectors, cosine compares their shape: zero means no overlap and one means identical direction. Mixture recovery ignores unmatched fitted topics, so it must be read alongside word recovery. Document completion tests prediction: infer topics from half the physical peak groups and score the other half, keeping a fragment and its loss together. This scores inferred mixtures rather than integrating posterior uncertainty. Lower plug-in negative log-likelihood (NLL), in nats per withheld in-vocabulary pseudo-token, is better. It is token-weighted and comparable across models on the same benchmark, not across the synthetic and real datasets. The real analysis scores 1,277,983 tokens from 3,888 eligible spectra; 3.14% of withheld tokens are out of vocabulary and excluded.

The entropy-effective topic count describes how many equally weighted topics would have the mixture’s entropy; it is one for a single topic and K for a uniform mixture. We report the median within each fit, then average across fits. Counting distinct topics that dominate at least one spectrum separately checks whether local compactness accompanies broad corpus-wide topic use. Compactness alone is not evidence of better chemistry.

Chemical screening with MAG and SOS. Mass2Motif Annotation Guidance (MAG) retrieves reference spectra for a topic and constructs a consensus molecular fingerprint [2]. Binary MACCS (Molecular ACCess System) fingerprints encode predefined molecular features. MAG operates after training. The annotation library excludes molecules sharing connectivity with validation or test compounds. Each validation spectrum is associated with its largest-weight topic, and repeated compounds are deduplicated within that topic.

The substructure overlap score (SOS) averages the fraction of the consensus fingerprint features present in each associated molecule. Four of five features gives a contribution of 0.8; it does not mean that 80% of the molecule’s atoms were identified. An *evaluable* motif survives MAG refinement, has a successfully constructed consensus and has at least one associated molecule with a valid fingerprint. A valid empty consensus scores zero; a missing consensus is not evaluable. A *useful* motif additionally has $\text{SOS} \geq 0.6$, the MS2LDA screening convention. Mean SOS includes only evaluable motifs, which can be different populations for different models. We therefore report evaluable counts, useful counts and mean SOS together. The detailed MAG protocol and SOS equation are in Supplementary Section S4. Neither this cutoff nor fingerprint containment establishes a chemically confirmed substructure.

3.9 Chemical specificity and correspondence

We assess whether the apparent chemical agreement exceeds that expected from common molecular features. For each motif, background SOS is the expected overlap among compounds matched by acquisition profile and precursor-mass band; excess SOS is observed minus background overlap. We also test enrichment of explicit MACCS molecular features and count distinct supporting compounds. Single-compound support is not evidence of recurring chemistry. Multiplicity-adjusted values $q \leq 0.05$ are exploratory screening criteria, not confirmation of discoveries. Null distributions, adjustment families and sensitivity protocols are specified in Supplementary Section S5.

These post-hoc analyses use the same three selected-model and three Tomotopy fits without retraining. Plain ETM is omitted because full arrays for one fit are unavailable. The protocol was fixed before these calculations, but the development-validation data had already informed model selection. The analyses therefore provide further characterization, not independent confirmation.

We also compare spectral repeatability across fits, overlap among supporting compound sets, and similarity to annotated MotifDB references. Chemical-effect cosine compares profiles of enriched or depleted MACCS features; supporting- compound Jaccard measures shared compounds relative to the union. Spectral agreement can coexist with different compound assignments. MotifDB similarity provides partial corroboration and is not a chemical identity probability. Matching, deduplication and mass-tolerance rules are in Supplementary Section S5.

3.10 Recovery between model families

To ask whether the models recover the same inventory, we find each source motif’s closest counterpart in a target fit and report the fraction reaching each cosine-similarity threshold. We evaluate both directions and compare cross-model curves with repeat-run agreement. Target motifs may be reused, and every source motif remains in the denominator, including weak matches. Cosine similarity is not a percentage of chemically identified motifs.

We consider all topics and a recurring, MAG-evaluable cohort supported by at least two compounds. Both inventories are filtered before matching. Sensitivity analyses remove the MAG requirement or give fragments and losses equal weight. Target-fit results are averaged within each source fit; means and sample SDs then summarize three source fits. These summaries share fitted models and are not independent chemical replicates. Example spectra follow a fixed score-quantile selection rule. Full equations, aggregation and selection rules are in Supplementary Section S6.

4 Results

4.1 Synthetic spectra: controlled recovery of planted motifs

At $K = 128$, the chosen model recovers an average of 15.67 of 18 planted motifs at the stated cosine criterion, compared with 9.00 at $K = 36$ (Table 1). Both word and mixture recovery improve with the larger number of fitted topics, while the median effective topic count remains near two. The model can therefore fit more topics than were planted while using compact mixtures in this simulator, although recovery is incomplete. These are frozen historical fits. A later correction to the gradient at uniform contextual evidence is provably inactive for these $K = 128$ fits, but its inactivity has not been established for $K = 36$ (Supplementary Section S8).

Table 1: Synthetic validation results for the selected model: mean $\pm$ sample SD over three data-generation/training repetitions, with 160 validation spectra per realization. Word and mixture recovery are the matched cosines defined in Section 3.8; higher is better. Recovered motifs are counted out of 18 at word cosine ≥ 0.50 . NLL is in nats per withheld pseudo-token; lower is better. Effective counts are within-fit medians.

Fitted K	Word recovery	Mixture recovery	Recovered / 18	NLL	Median K_{eff}
36	0.459 ± 0.041	0.724 ± 0.029	9.00 ± 1.73	6.279 ± 0.012	2.053 ± 0.079
128	0.630 ± 0.015	0.915 ± 0.015	15.67 ± 0.58	6.193 ± 0.055	1.716 ± 0.236

At $K = 128$, the mixture-recovery cosine is 0.915 whereas word recovery is 0.630. These scores measure different aspects of recovery and are not interchangeable accuracy estimates. Recovering relative use of the matched motifs does not imply equally accurate emissions, nor does the aligned mixture metric penalize all unmatched topic mass. These experiments assess recovery when the planted motifs are known. They do not establish a chemical advantage over Tomotopy on real spectra.

4.2 MSⁿLib: comparison with Tomotopy and the ETM base

Primary comparison with Tomotopy. The selected model yields 390.7 ± 2.1 useful motifs versus 351.7 ± 7.0 for Tomotopy, and 665.7 ± 7.6 evaluable motifs versus 578.0 ± 5.0 (Table 2). The difference in mean useful-motif count is +39.0 under the same chemical association rule. The increase means that more motifs meet the stated chemical screening criterion.

Table 2: Primary MSⁿLib validation comparison: $K = 1,000$ and 3,889 spectrum–topic associations per fit. Every metric is mean $\pm$ sample SD over three fitted runs on fixed data. Bold marks the best observed mean for directional metrics, not statistical significance. Plain ETM is the secondary enhancement baseline, not a replacement for the primary Tomotopy comparison. Evaluable and useful counts measure inventory breadth; larger is better under the stated definitions. Mean SOS is conditional on evaluable motifs; larger is better. NLL is in nats per withheld pseudo-token; lower is better. Effective counts are within-fit medians and are descriptive, not ranked. All metrics are defined in Section 3.8.

Model	Fits	Evaluable	Useful	Mean SOS	NLL	Median K_{eff}
Tomotopy LDA	3	578.0 $\pm$ 5.0	351.7 $\pm$ 7.0	0.639 $\pm$ 0.003	9.693 $\pm$ 0.004	9.26 $\pm$ 0.09
Plain ETM	3	202.7 $\pm$ 5.5	125.0 $\pm$ 6.1	0.637 $\pm$ 0.006	8.707 $\pm$ 0.029	40.75 $\pm$ 1.09
Contextual Sparse ETM	3	665.7 $\pm$ 7.6	390.7 $\pm$ 2.1	0.629 $\pm$ 0.002	9.525 $\pm$ 0.023	3.72 $\pm$ 0.03

Tomotopy has higher mean SOS among its evaluable motifs (0.639 versus 0.629), whereas the selected model has lower completion NLL (9.525 versus 9.693). The neural model therefore does not dominate Tomotopy on every measure. Mean SOS is calculated over a different set of evaluable motifs for each model and does not account for how many motifs remain unevaluable. The median entropy-effective topic count, averaged over the three fits, is 3.72 topics for the selected model and 9.26 for Tomotopy, indicating more concentrated neural mixtures under this diagnostic. Exact zero weights and chemically correct topic assignments are separate properties.

Comparison with plain ETM. Across three fits, plain ETM produces an average of 202.7 evaluable and 125.0 useful motifs, compared with 665.7 and 390.7 for the selected model. Its median entropy-effective topic count, averaged over the three fits, is 40.75 rather than 3.72. The enhanced model thus produces more useful motifs and more concentrated mixtures than the ETM baseline under these training settings. Plain ETM has better completion NLL (8.707 versus 9.525) and slightly higher conditional mean SOS (0.637 versus 0.629). These results do not establish that each enhancement is individually necessary. Differences in batch size and in the operating environment of one plain-ETM fit also limit the comparison.

Compact mixtures without corpus-wide collapse. All three real fits retain similar useful-motif counts and compact mixtures as summarized in Table 2. The median number of topics with strictly positive weight is six in every fit. The dominant-topic count is 807.3 $\pm$ 9.1 across fits: hundreds of distinct topics dominate at least one validation spectrum. Thus the observed local sparsity is not simply a consequence of using only a handful of topics across the entire corpus.

Similar aggregate scores do not show that every run discovers the same chemical motifs. The following analyses examine chemical specificity, supporting compounds and correspondence between the fitted inventories.

4.3 Agreement and differences with Tomotopy

The two models recover a shared subset of spectral patterns, but their motif inventories are not interchangeable. Directed recovery makes this distinction explicit: it measures how much of each source inventory has a close counterpart, without requiring a one-to-one mapping or omitting weak matches. The comparison is a central complement to the aggregate performance results, because similar motif counts and SOS values need not describe the same fragmentation patterns.

4.3.1 Recovery relative to repeat-run agreement

For each directed fit pair, we take the median closest-counterpart cosine over the complete source inventory. Averaging these medians over target fits and then source fits gives 0.454 from neural motifs to Tomotopy and 0.454 in the reverse direction. Both are below their repeat-run benchmarks, 0.650 for the neural model and 0.547 for Tomotopy (Table 3). Among recurring, evaluable motifs, these cross-model summaries are 0.420 and 0.460. Figure 4 shows the complete recovery curves, including a tail of close counterparts as well as the motifs with little correspondence. Tomotopy-to-neural recovery for the recurring, evaluable cohort lies closer to Tomotopy’s repeat-run benchmark than the reverse comparison lies to the neural benchmark. This is useful evidence of continuity, but not evidence that both methods recover the same inventory.

Table 3: Directed closest-counterpart agreement. Neural denotes the selected Contextual Sparse ETM. “All” and “Recurring” are full- β cosines; the latter filters both inventories to recurring, MAG-evaluable motifs before matching. Compound Jaccard and chemical-effect cosine use those recurring matches. Entries are mean $\pm$ sample SD over three source-fit summaries, each averaging the median across source topics over the two within-model or three cross-model target fits. Thus neither the SD nor the nine cross-model pairs represent independent chemical replicates. Every source topic remains in the recovery denominator; undefined chemical profiles are excluded only from chemical summaries. Metrics are defined in Section 3.10; the descriptive values are not ranked.

Source $\rightarrow$ target	All cosine	Recurring cosine	Compound Jaccard	Chemical cosine
Neural $\rightarrow$ neural	0.650 $\pm$ 0.003	0.625 $\pm$ 0.019	0.171 $\pm$ 0.005	0.423 $\pm$ 0.016
Neural $\rightarrow$ Tomotopy	0.454 $\pm$ 0.003	0.420 $\pm$ 0.006	0.075 $\pm$ 0.010	0.260 $\pm$ 0.013
Tomotopy $\rightarrow$ neural	0.454 $\pm$ 0.001	0.460 $\pm$ 0.011	0.089 $\pm$ 0.001	0.307 $\pm$ 0.010
Tomotopy $\rightarrow$ Tomotopy	0.547 $\pm$ 0.002	0.525 $\pm$ 0.015	0.167 $\pm$ 0.000	0.382 $\pm$ 0.009

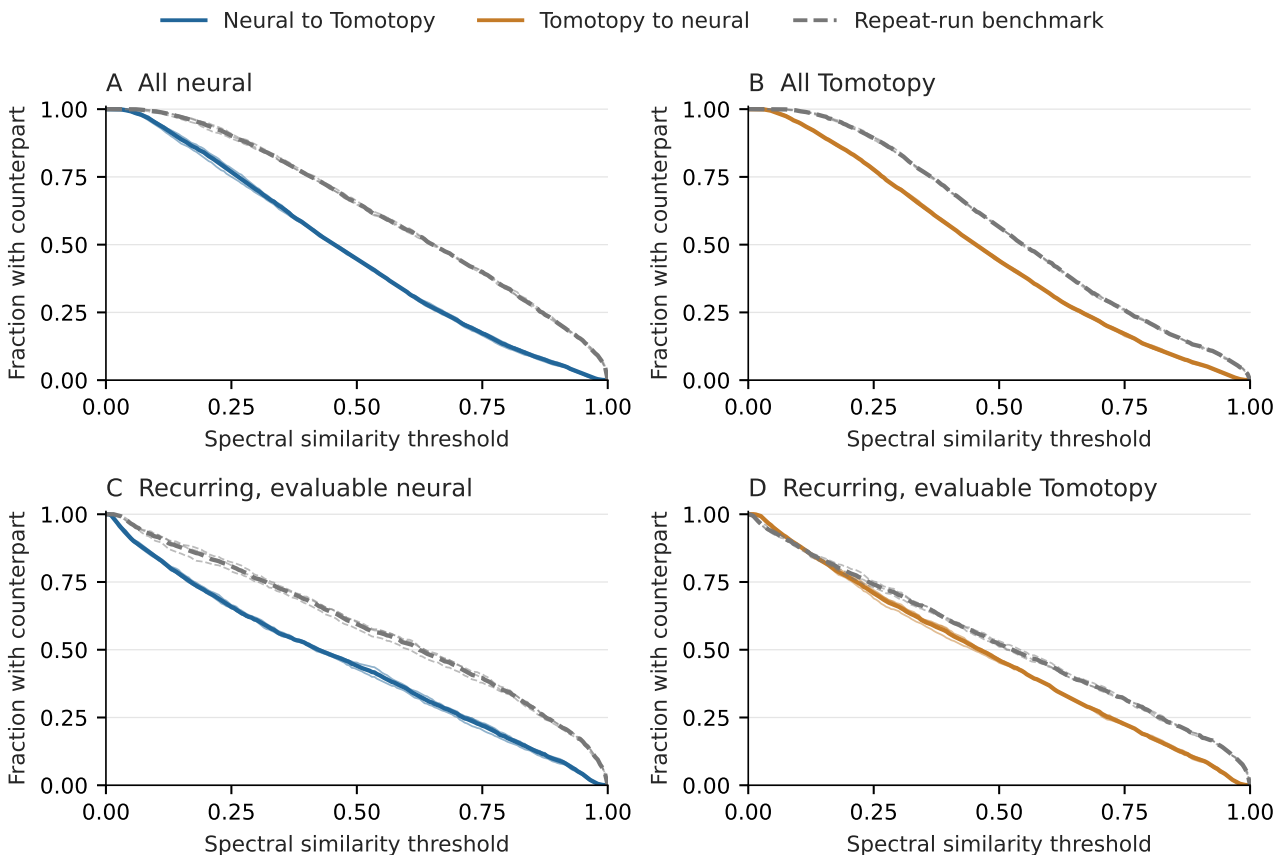


Figure 4: Bidirectional recovery relative to repeat runs. Colored curves show neural-to-Tomotopy (blue) or Tomotopy-to-neural (orange) recovery; grey dashed curves use other fits of the source model. Panels A–B include all 1,000 topics; C–D include recurring, evaluable topics, averaging 514.7 neural and 502.7 Tomotopy motifs per fit. The horizontal axis is the full- β similarity required of a counterpart; the vertical axis is the fraction of the entire source cohort meeting it. Thin lines are three source-fit summaries, each averaging target fits, and thick lines are their mean. They are not confidence intervals. No target fit is pooled with another and no threshold is treated as a chemical-correctness boundary.

The cohort definition changes the question as well as the number of motifs. Requiring recurrence without a MAG annotation gives cross-model median-score summaries of 0.516 and 0.403, respectively. Annotation availability therefore affects which patterns enter the comparison. In particular, an analysis restricted to successful matched pairs can look more similar than recovery of an entire eligible inventory. Here weak and unmatched source motifs remain visible in every curve.

4.3.2 Shared fragments do not imply identical chemistry

Agreement is much stronger in fragments than neutral losses for the closest recurring, evaluable cross-model pairs. Mean fit-pair median fragment cosine is 0.550 from neural motifs to Tomotopy and 0.547 in reverse, whereas the corresponding loss cosines are 0.006 and 0.007. Giving the two channel cosines equal weight reduces the median closest-counterpart score to 0.339 and 0.337 (Supplementary Figure S1D). High full-spectrum cosine can therefore reflect shared dominant fragments without close agreement across both channels. These observations depend on the selected cohort and matching rule; they do not establish which model’s neutral-loss patterns are chemically more accurate.

The matched chemical evidence is also partial. Compound Jaccard summaries are 0.075 and 0.089, with signed chemical-effect cosines of 0.260 and 0.307. The distributions in Supplementary Figure S1A–B show that higher spectral scores often accompany stronger feature agreement, but not uniformly. The two models can describe related fragment patterns while assigning them to different small compound groups. Agreement with Tomotopy is therefore corroboration for a shared pattern, not an independent structural annotation.

4.3.3 Concrete counterparts and one-to-many neighborhoods

The fixed example rule illustrates why a recovery curve is more informative than a single overlap percentage (Figure 5). The neural-to-Tomotopy pair nearest the median best score has cosine 0.422, with 1 directly shared fragment and 0 shared losses among its top words at 0.01 Da tolerance. The pair nearest the upper decile has cosine 0.887, but still shares only 2 top fragments and 0 top losses. Its dominant fragment aligns closely, while the remaining spectral description differs. Both are spectral probability patterns, not experimentally confirmed fragmentation assignments.

Allowing multiple counterparts avoids concealing possible splits or merges, but does not establish that they are common at high similarity. The source chosen near the upper decile of second-neighbor scores has closest scores 0.522 and 0.324. It shares 5 and 3 top fragments with the two Tomotopy candidates, respectively, but 0 top losses with either. The fraction with multiple counterparts falls rapidly as the required similarity increases (Supplementary Figure S1C). Thus the present evidence supports partial, sometimes one-to-many spectral correspondence rather than a near-identical chemical inventory. This provides a concrete basis for comparing the neural model’s mixture compactness and inference architecture with the established MS2LDA workflow, without assuming chemical equivalence.

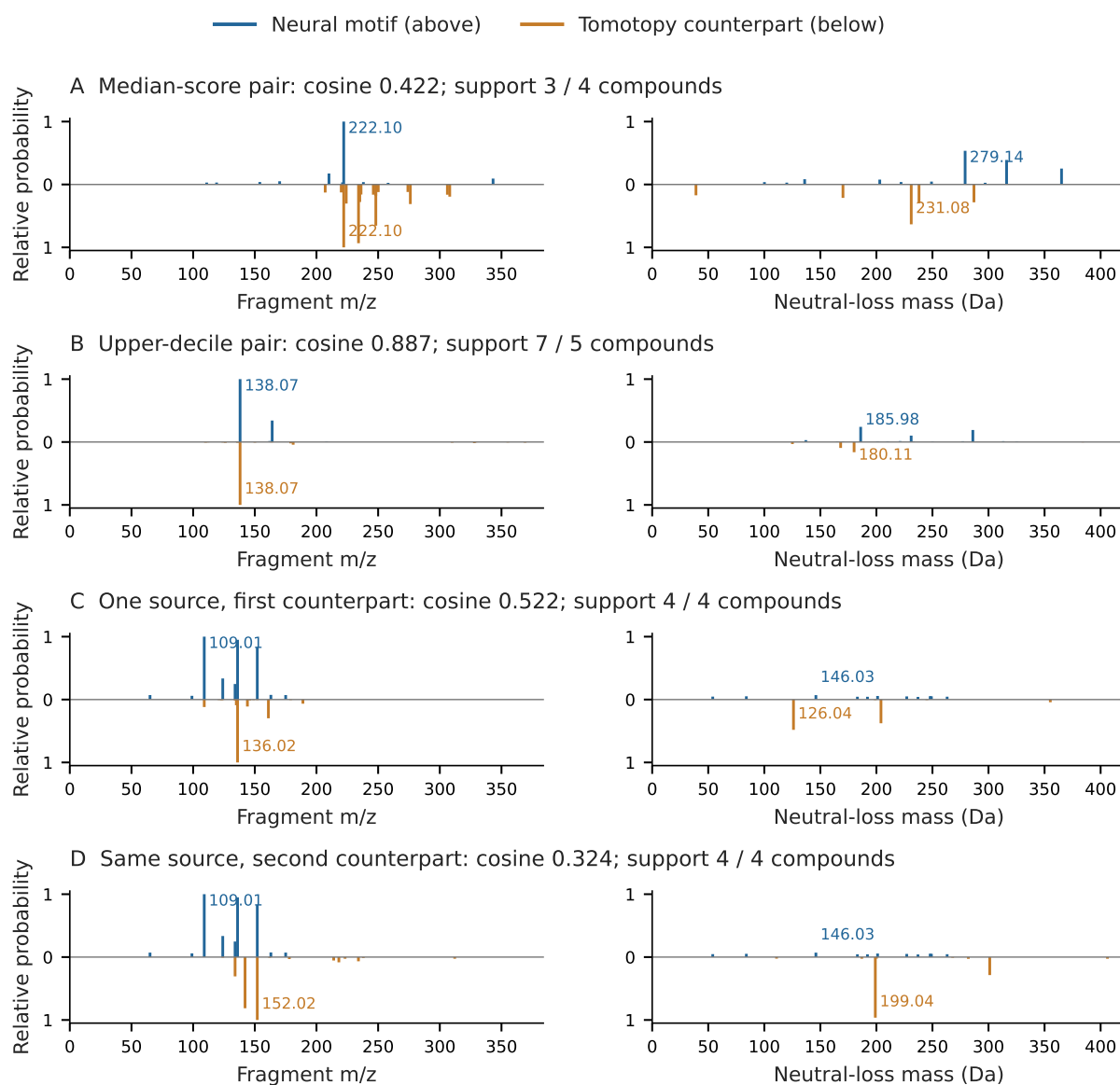


Figure 5: Actual motif counterparts selected by the fixed score-quantile rule. Neural top-20-word probabilities are plotted above zero and Tomotopy probabilities below zero, separated into fragment and neutral-loss channels. Each topic is divided by its own largest word probability across both channels; these are not measured ion intensities. Axes are shared in mass range within each column, and labels identify the largest displayed word in each channel. Support counts are distinct compounds, neural first. (A) Pair nearest the median closest-counterpart score. (B) Pair nearest its 90th percentile. (C–D) The same neural motif and its two nearest Tomotopy counterparts, chosen near the 90th percentile of second-best scores. The common source in C–D makes the one-to-many relationship visible without a graph of topic labels. Exact identities, all probabilities and mass-tolerance sensitivities are retained in the evidence. Examples are illustrative, not selected for chemical agreement and not representative chemical validations.

4.4 Chemical specificity beyond common molecular features

Both models have a small positive SOS excess over their matched backgrounds, but the raw overlap is much larger than this excess (Table 4; Figure 6A). The selected model’s mean background SOS is 0.594, compared with raw mean SOS 0.629; the corresponding Tomotopy values are 0.604 and 0.639. Both therefore have mean excess near 0.035. Details of the matched background are in Supplementary Section S5.1. Much of their fingerprint containment can be explained by common features among compounds with comparable acquisition profiles and precursor masses. This does not erase the observed associations, but it narrows what a high raw SOS can establish about chemical specificity.

Table 4: Post-hoc chemical assessment on the fixed MSⁿLib validation split. Entries are mean $\pm$ sample SD over three fits per model. “Evaluable, ≥ 2 ” counts MAG-evaluable motifs supported by at least two distinct compounds. Background SOS is expected overlap among acquisition/mass-matched compounds; excess SOS is observed minus background overlap. Both are averaged over evaluable motifs within a fit. “SOS screen” counts motifs with conditional SOS $q \leq 0.05$; “Feature screen” counts motifs with at least one MACCS key meeting that cutoff, regardless of MAG availability. Families contain all six fits. Counts are exploratory screens, not numbers of chemically confirmed discoveries. These descriptive metrics are not ranked. Definitions and caveats appear in Section 3.9.

Model	Evaluable, ≥ 2	Background SOS	Excess SOS	SOS screen	Feature screen
Tomotopy LDA	502.7 $\pm$ 13.0	0.604 $\pm$ 0.002	0.035 $\pm$ 0.001	9.7 $\pm$ 3.5	12.0 $\pm$ 3.0
Contextual Sparse ETM	514.7 $\pm$ 14.2	0.594 $\pm$ 0.003	0.035 $\pm$ 0.003	10.7 $\pm$ 1.5	10.0 $\pm$ 1.7

Support explains an important part of the inventory difference. Among MAG-evaluable motifs, 151.0 in the selected model and 75.3 in Tomotopy are supported by only one compound, averaged over fits. Requiring at least two compounds reduces the evaluable inventories to 514.7 and 502.7, respectively. The larger neural inventory is therefore not equivalent to a similarly large increase in recurring, independently supported chemistry. Figure 6B shows the full support distribution, including supported motifs without a MAG annotation.

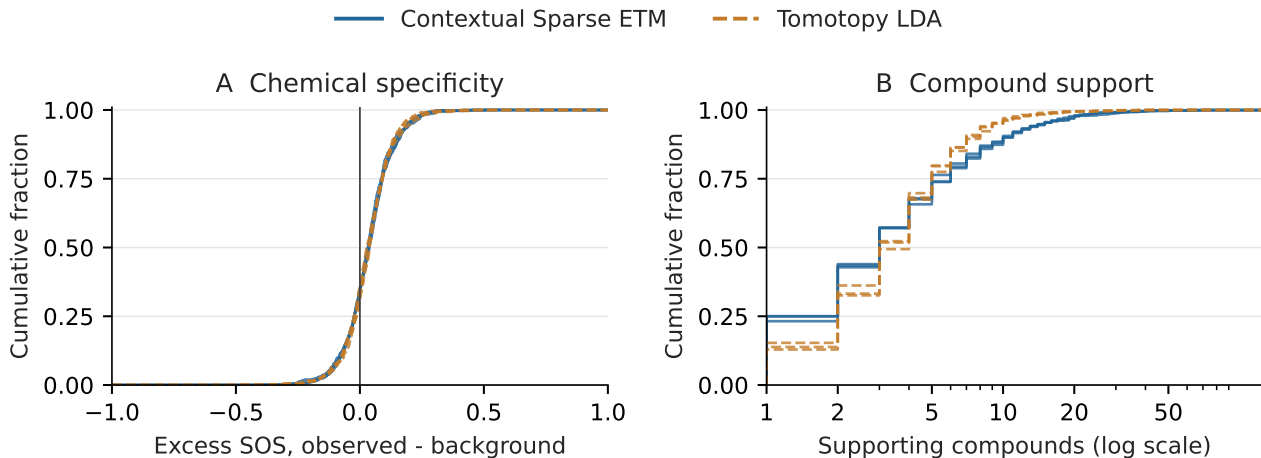


Figure 6: Chemical specificity and compound support, with one curve per fit. (A) Empirical cumulative distribution of SOS excess over the 50 Da acquisition-matched background, among evaluable motifs (657–671 per selected fit; 573–583 per Tomotopy fit). The vertical line denotes no excess. (B) Cumulative distribution of distinct-compound support among all supported motifs (799–817 selected; 933–959 Tomotopy), on a logarithmic count axis. These curves describe different motif inventories, not paired chemical identities or independent observations for significance testing.

Only a small subset passes the conservative exploratory screens: 10.7 versus 9.7 motifs per fit for SOS, and 10.0 versus 12.0 for at least one enriched MACCS feature. The latter tests are independent of MAG availability, not statistically independent of the SOS analysis. Detected features include bromine presence (key 46) in both models and sulfur in a ring (key 36) in selected-model motifs. These are interpretable structural associations, but a key does not specify a fragmentation pathway or a uniquely identified molecular substructure.

The small positive mean excess persists with alternative mass-bin widths of 25 and 100 Da (Supplementary Section S5.1). In the scaffold-balanced analysis it is 0.032 for the selected model and 0.029 for Tomotopy. The

feature screen then retains only 3.0 and 3.0 motifs per fit, respectively. That reduction reflects changed structural composition and reduced support; it does not by itself distinguish confounding from loss of power. Together, the analyses support some chemical structure in both inventories, without establishing a neural advantage in specificity.

4.5 Spectral repeatability does not imply identical compound groups

The selected model has higher within-model spectral repeatability than Tomotopy in these saved fits. The mean of the three pairwise median full- β cosines is 0.625 for the selected model and 0.529 for Tomotopy (Figure 7A). The mean reciprocal-nearest fraction is 0.673 versus 0.666. Across models, the mean of nine pairwise median cosines is lower, at 0.404: their inventories are not simply relabelled copies of one another. These descriptive pair summaries are correlated because they reuse the same fits. Neural repeats share the same fixed word embeddings, so this repeatability is conditional on that representation; sensitivity to re-learning the embeddings is not measured.

The supporting molecules are less repeatable than the spectral patterns. The mean within-model median compound Jaccard is 0.012 for the selected model and 0.000 for Tomotopy. Different runs can assign similar spectral motifs to different compounds, particularly when each motif has few dominant-topic associations. Feature-effect profiles and top-word overlaps are provided alongside these matches in the supporting evidence; neither forced matching nor similar spectra establishes identical chemistry.

Within-fit redundancy also cautions against reading motif counts as counts of distinct substructures. Nearest-topic similarity and repeated MAG fingerprints are reported in Supplementary Section S7.

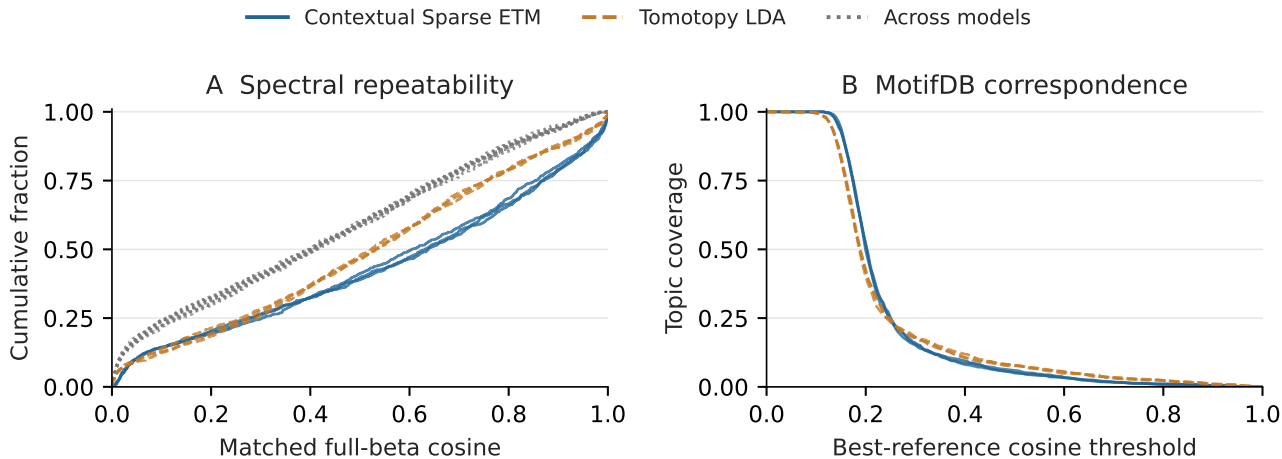


Figure 7: Spectral correspondence across fits and to annotated MotifDB references. (A) Each curve shows all 1,000 Hungarian-matched topic pairs for one fit pair: three selected-model pairs, three Tomotopy pairs and nine cross-model pairs. A curve farther right indicates higher matched cosines; the matches include weak forced assignments. (B) Fraction of all 1,000 topics per fit whose best-reference Spec2Vec cosine reaches each threshold. The two panels use different representations: full topic-word probabilities in A and embedded top-20-word pseudo-spectra in B. Thresholds in B are displayed continuously, not interpreted as chemical correctness probabilities.

4.6 MotifDB provides partial corroboration for selected examples

The 255 reference entries reduce to 234 distinct typed spectral signatures, all scorable with the fixed Spec2Vec model. No reference is excluded as malformed; 8 deduplicated signatures retain different manual-label strings. All learned topics also have scorable embeddings. Most matches are modest: the mean within-fit median best-reference cosine is 0.201 for the selected model and 0.188 for Tomotopy. The coverage curves cross (Figure 7B), so the small median difference is not a uniform neural advantage. At 0.01 Da, the best reference directly matches an average fraction 0.066 of the 20 query words for the selected model, compared with 0.069 for Tomotopy.

Table 5: Illustrative selected-model matches: the three highest-cosine, distinct reference signatures among recurring motifs with nonzero direct overlap. These are not a representative sample. Reference labels are shortened for display, not assigned as confirmed identities of the learned motifs. "Compounds" is distinct motif support; "Cosine" is Spec2Vec similarity. Fragment and loss columns show matched/query feature counts at 0.01 Da. Examples 2 and 3 match distinct reference signatures carrying the same structural description. Full fit/topic IDs, labels, source IDs, reference denominators and tolerance sensitivities are retained in the generated evidence summary.

Example	Reference description	Compounds	Cosine	Fragments	Losses
1	Fragment indicative for aromatic compounds related to methylbenzene substructure	36	0.950	1/4	0/16
2	Fragments indicative for dihydroxylated benzene ring substructure	19	0.945	2/3	0/17
3	Fragments indicative for dihydroxylated benzene ring substructure	20	0.942	2/2	0/18

The strongest examples illustrate both the value and the limits of reference matching. An aromatic-fragment reference can closely resemble the dominant part of a motif while explaining only one of its twenty words; the dihydroxylated-benzene examples share two fragment masses but no losses (Table 5). High embedding similarity therefore need not mean complete motif agreement. These matches offer hypotheses for inspection, not confirmation that each annotated substructure is present in every supporting molecule. Weak reference correspondence likewise cannot be used to claim novel chemistry.

5 Discussion

5.1 Discovery breadth, chemical specificity and predictive fit

Our model produces more motifs meeting the usefulness criterion than Tomotopy on the same validation data (Table 2). The difference in mean useful-motif count is +39.0, providing more candidates for chemical interpretation. Additional motifs need not represent additional unique substructures, however. Several motifs may describe related chemistry or different fragmentation behavior of the same substructure, so a larger number of motifs does not necessarily provide more independent chemical information.

The compound-level assessment sharpens this distinction. Both models show a small positive SOS excess after accounting for common fingerprint features within matched acquisition/mass groups, with no clear difference in average excess. Moreover, much of the larger evaluable neural inventory consists of motifs supported by a single compound. The gap becomes much smaller when recurrence across compounds is required. Thus the main evidence favors broader candidate generation, not a proportional increase in chemically specific, recurring substructures. The enriched MACCS features and selected reference matches provide useful starting points for interpretation, but their restricted support and overlapping descriptions prevent a direct count of unique chemical discoveries.

The directed comparison with Tomotopy is an important qualification to this discovery claim. It demonstrates shared spectral patterns, including close counterparts, while retaining the source motifs that do not match well. Recurring Tomotopy motifs are partly recovered by the neural model, but cross-model recovery remains below within-model repeatability. Fragment agreement is much stronger than neutral-loss agreement, and supporting compound groups often differ. The result is evidence of partial continuity with established MS2LDA outputs, not a claim that two implementations discover the same chemical substructures. Neither agreement nor disagreement alone decides whether a motif is chemically valid.

Repeatability adds a different consideration. Neural spectral patterns match more closely across runs, while dominant-topic compound groups remain unstable in both models. Spectral repeatability is useful for selecting motifs to inspect, but is not the same as stable assignment of a molecular class. The repeated MAG profiles and partial reference overlaps further show that a motif inventory contains overlapping levels of description, from broad chemical features to more specific fragment patterns.

Predictive fit reveals a further trade-off. Completion NLL is lower than Tomotopy’s but higher than plain ETM’s, even though plain ETM produces far fewer useful motifs. Plain ETM also has slightly higher conditional mean SOS than the selected model. The preferred model therefore depends on whether priority is given to word prediction, conditional SOS, or useful-motif count and compact assignments. Count likelihood and MAG/SOS measure different properties: spectral-word prediction and library-dependent structural agreement. They should be assessed separately, and sparse mixtures alone are not sufficient reason to accept poorer predictions.

5.2 Compact mixtures and the ETM foundation

Local sparsity can coexist with broad use of topics across the corpus. The selected model has a median entropy-effective count of about 3.7 topics per spectrum, while about 807 topics dominate at least one validation spectrum. Entmax permits exact zero topic weights, and the fitted model uses many different topics despite its compact per-spectrum mixtures. The synthetic experiments show a similar pattern when the planted motifs are known. However, the gap between word and mixture recovery shows that accurate relative assignments need not imply equally accurate spectral patterns. Mixture size and topic counts should therefore be interpreted alongside recovery and chemical evaluation, rather than judged against a single threshold.

The model builds on the published ETM, retaining its embedding-based topic-word decoder, Gaussian latent variables, MLP inference network and latent-space variational objective. The enhancements modify three parts of this model: balanced channels separate fragment and loss normalization, contextual evidence shifts the Gaussian posterior mean using the shared embedding geometry, and entmax changes the induced topic-mixture distribution. The resulting topics remain normalized word distributions that can be inspected as spectral motifs. The MLP is retained from ETM rather than introduced as a new component.

The plain-ETM comparison assesses the combined enhancements rather than each component separately. Earlier ablations in related model variants provide more limited evidence for particular mechanisms, as summarized in Supplementary Section S8.3; they do not test every combination of enhancements in the final model. Together, these results support the combined ETM extension and document its trade-off between motif discovery and predictive fit. They do not establish that every component is necessary in all settings or that no simpler model could achieve similar results.

5.3 Limitations of the current evidence

We used the same validation data to select the model and report its performance. Repeated fits describe training variability on those data, but do not remove selection bias or show how performance varies across new chemical populations. The reserved test data were also used during earlier development and therefore cannot provide independent confirmation of the selected model. Three fits per model do not establish statistical significance. In the plain-ETM comparison, the original batch size is retained and one fit used a different operating environment. That comparison does not fully separate architectural effects from optimization and runtime differences, even with fixed data and matching scientific package versions.

The observation model introduces further restrictions. Pseudo-counts, two-decimal bins and a fixed vocabulary make spectra compatible with a topic likelihood, but fragment and loss words from the same peak are dependent even though the multinomial treats token draws as conditionally independent. The pseudo-count scale also fixes the balance between reconstruction and KL regularization. Equal channel mass and the sparse Gaussian-derived prior impose additional assumptions that may not suit every dataset. Similarly, recovery in one 18-motif simulator shows behavior under its construction, rather than validating real fragmentation mechanisms or robustness across all spectral settings.

Chemical evaluation is constrained by its reference library and association rule. Connectivity filtering prevents an evaluated molecule from serving as its own annotation evidence, but cannot remove all related structures or acquisition biases. Assigning each spectrum to its dominant topic provides a common comparison rule while excluding secondary motifs from chemical scoring. The SOS cutoff defines a screening criterion rather than validating a molecular substructure. The new conditional nulls make that limitation more explicit, but cannot eliminate residual dependence among related structures or guarantee exchangeability within strata. Their multiplicity-adjusted values are exploratory because the model was selected on these data. Scaffold balancing reduces overrepresentation but also reduces statistical power. Finally, MotifDB coverage is limited and its Spec2Vec matching is partly correlated with MAG. These automated checks can prioritize motifs without expert input; none confirms a fragmentation mechanism or a novel substructure on its own.

Cross-model matching has its own limits. A closest counterpart is the maximum over many candidates and can share a dominant peak without the remaining pattern. Candidate-set sizes, annotation availability and channel weighting all affect recovery, which is why we retain complete denominators and report the sensitivities. One-to-many spectral neighbors can reflect overlapping features rather than a clean chemical subdivision. Moreover, comparison with Tomotopy does not create an independent reference standard: both models use the same spectra, vocabulary and downstream annotation resources.

5.4 Implications for validation and future encoders

To assess how well these findings generalize, the model should be compared with Tomotopy on data not used for model development or selection. Such a comparison should set the evaluation criteria in advance and use comparable training and inference conditions for both models. The present automated assessment provides a reproducible way to select motifs for that next study: consider effect above background, support across

compounds and scaffolds, repeatability, and agreement with concrete reference features together. The next chemical step is to inspect the supporting spectra and test the proposed structural relationships, rather than attach a definitive label from a high embedding score. Independent molecules and, where available, targeted standards or expert structural assessment would help distinguish new chemical information from alternative descriptions of shared chemistry.

The partial agreement also clarifies which practical advantages should be tested next. Tomotopy can infer topics for previously unseen spectra; the neural distinction is reuse of a learned inference function rather than iterative document-specific inference. Latency, throughput and memory should therefore be compared at comparable predictive and motif quality before claiming a computational advantage. More compact mixtures and greater spectral repeatability are observed here, but neither automatically implies faster inference or superior chemical annotation.

The Gaussian inference network also provides a way to incorporate richer spectral representations. A future encoder could predict Gaussian means and log variances directly from spectra, or from DreaMS representations, while retaining the current topic decoder and its interpretation [16]. Such an extension would require separate evaluation. During document completion, neither the encoder nor the contextual pathway could use withheld peaks. Replacing the encoder would not eliminate tokenization from the complete model: its likelihood and motif representation would still depend on discrete spectral words. Modeling continuous spectra at the output would require a separate change to the observation model.

6 Conclusion

Contextual Sparse ETM adapts the published Embedded Topic Model through balanced fragment/loss emissions, whole-spectrum contextual posterior evidence and 1.5-entmax mixtures. It preserves an explicit topic–word representation and the neural Gaussian inference backbone while producing compact per-spectrum assignments alongside broad topic use across the corpus.

Across three training runs evaluated on the MSⁿLib validation split, the model produces more motifs meeting the usefulness criterion and lower completion NLL than Tomotopy, but lower mean chemical overlap among evaluable motifs. Automated chemical assessment finds modest background-adjusted agreement in both models, many single-compound candidates, and limited reference corroboration. Directed recovery identifies shared spectral patterns with Tomotopy but also substantial unmatched or weakly matched portions of both inventories. The closest cross-model counterparts agree more strongly in fragments than neutral losses, so this continuity does not establish chemical equivalence. The neural model’s greater spectral repeatability does not establish stable compound assignments or more unique substructures. These results support the model as a candidate-generation method with an inspectable ETM foundation, while keeping its chemical claims narrower than its motif counts. Independent evaluation and structural follow-up are needed to determine which candidates provide reproducible new chemical information.

Data and code availability

Public input and annotation assets are distributed in Zenodo record 20179680 [10]. Research code, manuscript sources and versioned evidence are available in the [public research branch of the MS2LDA fork](#). The implementation, evidence locations and reproduction settings are described in Supplementary Section S8 and the accompanying source inventory. The neural model remains an isolated research path; the production MS2LDA Tomotopy backend is unchanged.

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